DECISION	Start with a measurable Nesting Intelligence pilot - not a broad AI transformation

MANAGEMENT VIEW

1. Executive Summary

Client B is a structural steel fabricator processing approximately 80,000 tonnes per year across 10-15 concurrent projects. The client reports 7-10% overall material wastage, with plate wastage of about 20-25% and profile wastage of about 5-10%. Nesting is largely manual, while Strumis already provides an automatic nesting capability. The client wants evidence of where waste is avoidable and whether better decisions can reduce it.

Core FDE conclusion

The immediate problem is not "build AI for nesting." It is to establish a trusted material-utilisation baseline, objectively compare the current manual process with existing Strumis automation, identify the operational constraints that explain the gap, and then deploy a human-reviewed optimisation layer only where measurable value is proven.

Recommended outcome

- Phase 0: 1-2 week data and process audit to make waste, offcuts, revisions and nesting outcomes measurable.
- Phase 1: 6-8 week pilot that benchmarks manual nesting vs Strumis auto-nesting and a constraint-aware optimisation approach, including offcut reuse.
- Phase 2: production integration only after the pilot clears a predefined business and manufacturing gate.
- Treat drawing revision intelligence, yard optimisation, trailer planning and equipment utilisation as follow-on use cases, not part of the first build.

Why this is the right first move

Decision question	Why it matters	My current position
Is manual actually better than auto?	Existing automation is already available.	Benchmark before replacing or rebuilding it.
How much waste is avoidable?	Reported waste is not the same as recoverable waste.	Measure job-level losses before making savings claims.
What constraints are missing?	Manufacturing rules may explain operator preference.	Capture and encode them before optimisation.
Can offcuts be reused?	Reusable remnants may substitute for new stock.	Make this an explicit optimisation decision.

Business-case guardrail

No savings number is claimed at this stage. Annual value will be calculated from validated tonnes recovered multiplied by the client's actual material acquisition cost per tonne.

2. Current State, Problem Definition & Unknowns

End-to-end process observed

Stage	Current flow	Key issue / data opportunity
1. Commercial	Business Development -> Estimation -> Win	Project budget fixed at start; assumptions become operational constraints.
2. Engineering	Planning -> Engineering -> Fabrication drawings	Frequent revisions; loss attributable to engineering is not well measured.
3. Material	Purchase -> Project allocation -> Warehouse/Yard	Material is bought and allocated per project; storage sequence is imperfect.
4. Nesting	Manual nesting in current process; Strumis has auto-nesting	No trusted comparison of utilisation, scrap and planning time.
5. Production	CNC cutting -> fabrication -> coating	A technically efficient nest must still be executable on the shop floor.
6. Dispatch	Loading -> trailers -> delivery	Fleet utilisation is low; separate opportunity for later optimisation.

Problem statement

How can the client reduce avoidable material wastage without increasing manufacturing risk or replacing the existing Strumis workflow? The first release should improve the quality of nesting and reuse decisions while preserving operator control and an auditable path to CNC.

Known vs unknown

Known from discovery	Unknown / must validate
~80k tonnes/year; 10-15 projects at once.	Actual waste by nesting job, material, grade and thickness.
Plate waste ~20-25%; profiles ~5-10%; overall ~7-10%.	Why operators reject Strumis auto-nesting; configuration vs algorithmic limits.
Manual planning; Strumis/PLM in core workflow.	Exact CNC and manufacturing constraints used by nesting engineers.
Offcuts are recorded and sometimes checked for future jobs.	Offcut dimensions, quality, location and eligibility for reuse.
Drawing revisions are frequent.	Version-to-version material impact and who/what caused the change.
Material generally allocated to projects.	Whether cross-project reuse is allowed commercially/accountingly.

Priority questions

- P0: What historical nesting, material-consumption, offcut and scrap records can be exported from Strumis/PLM?
- P0: For the same historical job, can we obtain the manual nest, Strumis auto-nest and actual production outcome?
- P0: What shop-floor constraints make a mathematically better nest unacceptable?
- P1: Can offcuts be consumed across future jobs, and can material be shared across projects?
- P1: What is the client's fully loaded material acquisition cost per tonne for ROI calculations?

3. Options, Recommendation & What Not to Build

Options considered

Option	Value	Main risk	Decision
1. Analytics-only dashboard	Fast baseline; low change risk.	Does not improve the decision itself.	Use in Phase 0, not final solution.
2. Strumis configuration / workflow tuning	Leverages existing capability; potentially lowest integration effort.	May not address missing constraints or offcut strategy.	Benchmark first; pursue where evidence supports.
3. Custom optimisation layer around existing systems	Improves decisions while retaining existing ERP/PLM/CNC stack.	Requires constraint capture and integration.	RECOMMENDED.
4. Fully autonomous nesting-to-CNC	Highest theoretical automation.	Operational and safety risk; poor adoption if trust is low.	Do not build initially.
5. LLM-first "AI nesting"	Good for natural-language assistance.	LLMs are not the right primary engine for geometric optimisation.	Do not use as core solver.

Recommendation

Build a "Nesting Intelligence" decision layer around Strumis/PLM: ingest jobs, stock and offcuts; enforce manufacturing constraints; generate ranked nesting candidates; let the nesting engineer approve or modify the result; then pass the approved plan into the existing production workflow.

What the first pilot solves

- Nesting optimisation for plates and profiles using constraint-aware search / operations-research methods rather than an LLM.
- Offcut matching: identify reusable remnants before new material is selected.
- A/B benchmarking: current manual nest vs Strumis automatic nest vs proposed optimiser.
- Decision trace: show why a recommendation was generated, what constraints were active and what material was saved or consumed.

Explicit non-build list

- No ERP replacement; Strumis remains the system of record.
- No direct autonomous CNC control during the pilot.
- No cross-project material transfer automation until commercial/accounting rules are approved.
- No warehouse robotics / computer vision in the first release.
- No attempt to solve nesting, yard, trailers and equipment utilisation in one project.
- No savings guarantee before a historical baseline and controlled pilot exist.

Why this is more than an "AI model"

The core technical problem is constrained combinatorial optimisation: how to place required parts onto available stock while satisfying geometry, grade, thickness, grain, kerf/spacing, machine, handling and production constraints. ML can be added later for demand prediction, anomaly detection or ranking search strategies, but it is not necessary to replace the optimisation engine.

4. End-to-End Solution Engineering

Target architecture

Layer	Responsibility	Implementation direction
Source systems	Strumis, PLM, inventory/offcut records, production outcomes	Start with export/files; move to API integration only after data contract is stable.
Data layer	Jobs, parts, stock, offcuts, revisions, actual usage	PostgreSQL or equivalent; immutable job snapshots for auditability.
Constraint engine	Manufacturing rules and eligibility checks	Versioned rules; validated with nesting engineers.
Optimisation engine	Generate feasible candidate nests	Heuristics + search/constraint optimisation; compare with current Strumis auto-nest.
Offcut matcher	Find existing remnants that can replace new stock	Geometry + material compatibility + project eligibility.
Recommendation service	Score/rank candidates and expose expected impact	FastAPI service with traceable inputs/outputs.
User workflow	Review, modify, accept or reject	Simple web UI; human-in-the-loop by default.
Production handoff	Approved plan continues through current process	No direct CNC automation in pilot.
Evaluation/monitoring	Track utilisation, scrap, time, feasibility, acceptance	Dashboard + experiment log + audit trail.

Operational workflow

Step	System action	Human action
1	Import a new nesting job and eligible stock/offcuts.	Confirm job priority and constraints.
2	Validate material, geometry, grade, thickness and machine rules.	Resolve data exceptions.
3	Generate multiple feasible candidates.	Review visual result and operational fit.
4	Rank by material utilisation, scrap, offcut use and secondary objectives.	Accept, modify or reject.
5	Store approved plan and reasons.	Release through existing workflow.
6	Capture actual consumption and leftovers.	Close the loop for evaluation.

Integration principle

Use the client's existing systems as the source of truth wherever possible. A thin integration layer reduces implementation risk: Phase 0 can work from structured exports, while production integration is introduced only once the data contract, KPIs and user workflow are validated.

Future expansion path

Once nesting value is proven, the same operational data model can support drawing-revision impact tracking, warehouse location/sequence optimisation, trailer-load planning and equipment-utilisation optimisation.

5. Evaluation, Business Case & Go/No-Go Criteria

Evaluation design

Evaluation must happen before production deployment. Use a representative historical sample (target: at least 100 nesting jobs, stratified across plate/profile, grades, thicknesses and job complexity where data permits) and reproduce the same input information that was available at the decision time.

Metric	Definition	Pilot interpretation
Material utilisation	Required-part material / total material consumed	Primary KPI; improvement over manual baseline.
Scrap rate	Scrap / total material consumed	Lower is better.
Reusable offcut rate	Eligible offcut tonnes actually consumed / eligible offcut tonnes	Shows whether reuse logic is working.
Material saved	Baseline tonnes - pilot tonnes	Convert to value using actual acquisition cost.
Planning time	Time to produce an approved nest	Must not materially increase.
Feasibility failure rate	Recommendations rejected for production reasons	Must remain below agreed threshold.
Operator acceptance	Accepted recommendations / presented recommendations	Leading indicator of adoption.

Pilot comparison

Same job inputs	Manual baseline	Strumis auto	Proposed optimiser
Parts + quantities	Historical result	Re-run where supported	Generate candidates
Stock availability	Historical result	Use same stock set	Use same eligible stock
Constraints	Implicit in operator process	Existing configuration	Explicit, versioned rules
Outcome	Utilisation / scrap / time	Utilisation / scrap / time	Utilisation / scrap / time

Business-case sensitivity

Annual utilisation improvement	Recovered tonnes/year at 80,000 t/year
0.5 percentage point	400 t
1.0 percentage point	800 t
1.5 percentage points	1,200 t
2.0 percentage points	1,600 t

Annual material value = recovered tonnes x actual material acquisition cost per tonne. These are sensitivity calculations, not claimed savings. The 7-10% reported waste range should not be treated as fully preventable.

Go / no-go gate

- Go to production only if the pilot demonstrates a repeatable utilisation improvement (target: at least +1.0 percentage point aggregate, subject to data quality and job mix) without unacceptable feasibility failures or increased planning time.
- Go to a narrower use case if improvement is concentrated in specific plate/profile families or constraints.
- Stop or redesign if the optimiser cannot beat the manual baseline on a representative sample, or if the apparent gains depend on infeasible nests or data leakage.

6. Build Plan, Commercials, Risks & Dependencies

Implementation plan

Phase	Duration	Deliverables	Gate
0. Discovery + data audit	1-2 weeks	Data map, baseline, constraint catalogue, KPI definitions, integration plan, ROI model.	Is data sufficient and opportunity measurable?
1. Nesting Intelligence pilot	6-8 weeks	Benchmark harness, optimisation engine, offcut matcher, reviewer UI, evaluation dashboard.	Does pilot create validated value safely?
2. Production rollout	8-12 weeks	Strumis/PLM integration, monitoring, audit trail, workflow hardening.	Production readiness review.
3. Expansion	After proof	Revision intelligence, yard, trailer and equipment use cases.	Prioritise by ROI.

Indicative commercial model

Engagement	Indicative fee	Notes
Phase 0 - Discovery	₹3 lakh	Fixed fee; includes data audit and business case.
Phase 1 - Pilot	₹12 lakh	Fixed fee; conditional on Phase 0 data readiness.
Phase 2 - Production	₹20 lakh	Only after pilot gate; final quote depends on integration scope.
Total to first production	₹35 lakh	Indicative only; excludes taxes, third-party licence upgrades, hardware and major customisation outside agreed scope.

Suggested team

- 1 Forward Deployed Engineer / product lead: discovery, stakeholder alignment, delivery and ROI.
- 1 Optimisation / ML engineer: constraint modelling, benchmark and optimisation engine.
- 1 Data / backend integration engineer: data pipelines, API, database and Strumis/PLM integration.
- Part-time domain SME / QA: manufacturing validation and pilot acceptance.

Key risks and mitigation

Risk	Mitigation
Poor or incomplete historical data	Phase-0 quality gate; document gaps before modelling.
Tacit nesting knowledge not captured	Structured operator workshops; convert rules into versioned constraints.
Existing auto-nesting is configured poorly	Benchmark configuration separately from algorithmic capability.
Theoretical nests fail in production	Enforce machine/shop-floor constraints and require human approval.
Strumis integration takes longer than expected	Start with export/import; integrate APIs after the workflow is proven.
Cross-project reuse causes accounting conflict	Keep cross-project allocation out of pilot until approved.
Savings smaller than expected	Use explicit pilot gate and stop conditions.

Decision requested

Approve the 1-2 week discovery engagement. Commit to the pilot only after a quantified baseline and feasibility gate.

A1. Appendix: Ranked Questions & Assumptions

P0 assumptions to test

Assumption	Why it is needed	Validation method
Historical nesting data exists or can be reconstructed.	Without it, no trustworthy baseline or benchmark.	Strumis/PLM export + sample trace.
Manual operator decisions can be explained.	Optimiser must encode operational constraints.	Operator walkthroughs on real jobs.
Actual material consumption can be tied to nesting jobs.	Required to distinguish theoretical from realised savings.	Job/heat/stock genealogy check.
Strumis auto-nesting can be re-run or its outputs exported.	Core benchmark depends on it.	Run matched historical examples.
Offcuts are measurable and quality-eligible.	Needed for reuse optimisation.	Inventory audit and sample matching.

Detailed questions for first discovery with technical owners

1. What exact inputs are sent to the nesting module: part geometry, quantity, grade, thickness, stock size, grain and machine constraints?
2. What makes an otherwise efficient nest unusable: kerf, minimum spacing, handling, lifting, cutting sequence, remnant size, orientation, heat treatment or other rules?
3. Can the same released drawing set be linked to the material and nest version that was actually cut?
4. How are reusable offcuts uniquely identified, measured, located and marked as available/unavailable?
5. What rules govern project-level ownership, cost allocation and cross-project material consumption?
6. For each drawing revision, can we identify changed parts, changed material quantity and material already purchased/cut?
7. What is the current human nesting time per job and what percentage of recommendations would operators realistically review?

Required access / dependencies

- Historical nesting files/results, material consumption, scrap/offcut records and stock inventory.
- Strumis/PLM data dictionary or export specification; CNC and manufacturing constraints.
- Drawing revision history and material impact records where available.
- Procurement cost per tonne and agreement on the financial definition of savings.
- Working sessions with nesting, engineering, purchasing, warehouse/yard and production owners.

Assumption discipline

Any assumption that affects the algorithm, financial case or production workflow should be labelled and validated. Until validated, it must not appear as a claimed capability or guaranteed result.

A2. Appendix: Example Decision Logic & Calculation Notes

Illustrative recommendation score

A candidate nest can be ranked using a weighted objective, for example: material waste + offcut opportunity cost + planning effort + production penalties. The exact weights should be learned from operator decisions and agreed business priorities; they should not be hard-coded without validation.

Candidate	Utilisation	Offcut use	Feasibility	Planning time	Decision rationale
A	High	Low	High	Low	Strong baseline candidate
B	Very high	High	High	Medium	Potential preferred option
C	Highest	High	Low	Low	Reject despite high theoretical yield
D	Medium	Very high	High	High	Useful when offcut disposal risk is high

Financial model

- Recovered tonnes/year = annual processed tonnes x utilisation improvement in decimal form.
- Annual gross material benefit = recovered tonnes/year x actual material acquisition cost/tonne.
- Net pilot/production benefit = gross material benefit - software/service cost - incremental operating cost.
- Payback period = implementation cost / annualised net benefit.
- Do not monetise client-claimable revision losses or project-transfer savings until the accounting treatment is confirmed.

Prototype recommendation

A prototype is optional. If built, it should prove one specific point: for a small synthetic set of rectangular parts and stock sheets, a constrained optimiser can generate a feasible layout and compare it against a simple baseline. It should be labelled synthetic and should not be presented as evidence of client savings.

Data privacy

Client information should remain confidential. Any synthetic prototype data should be clearly identified as synthetic and used only to demonstrate workflow or algorithmic feasibility.

Appendix A3 — Effort, pricing and operating assumptions

Commercial estimates for planning only. They are not supplier actual costs or a binding quote; confirm scope and assumptions during discovery.

1. Team effort estimate

The proposed team is one FDE/product lead, one optimisation engineer, one data/backend integration engineer and part-time manufacturing SME/QA. The loaded daily rates below are assumptions used to make the effort estimate explicit.

Phase / role	Effort × assumed rate	Labour estimate
Discovery — FDE / product lead	6 days × ₹10,000	₹60,000
Discovery — data / integration engineer	4 days × ₹8,000	₹32,000
Discovery — optimisation engineer	3 days × ₹10,000	₹30,000
Discovery — manufacturing SME / QA	2 days × ₹7,000	₹14,000
Discovery total	**15 person-days**	**₹1.36 lakh**
Pilot — FDE / product lead	10 days × ₹10,000	₹1.00 lakh
Pilot — optimisation engineer	24 days × ₹10,000	₹2.40 lakh
Pilot — data / integration engineer	18 days × ₹8,000	₹1.44 lakh
Pilot — manufacturing SME / QA	8 days × ₹7,000	₹0.56 lakh
Pilot total	**60 person-days**	**₹5.40 lakh**
Production — FDE / product lead	12 days × ₹10,000	₹1.20 lakh
Production — optimisation engineer	20 days × ₹10,000	₹2.00 lakh
Production — data / integration engineer	30 days × ₹8,000	₹2.40 lakh
Production — manufacturing SME / QA	12 days × ₹7,000	₹0.84 lakh
Production total	**74 person-days**	**₹6.44 lakh**

The labour subtotal is not the same as the fixed fee. The fee also has to cover delivery management, requirements and documentation, QA, integration work, environment setup, contingency for fixed-scope delivery risk, handover and commercial margin. The exact allocation should be refined after discovery; the difference must not be described as pure profit.

2. Proposed fee and payment milestones

Stage	Fee	Payment milestones
Discovery and data audit	**₹3 lakh**	50% at kickoff (₹1.5 lakh); 50% on delivery of baseline, findings and go/no-go recommendation (₹1.5 lakh).
Nesting pilot	**₹12 lakh**	40% at start (₹4.8 lakh); 40% at benchmark/demo (₹4.8 lakh); 20% on pilot evaluation report (₹2.4 lakh). Subject to data readiness.
Production integration	**₹20 lakh**	30% at kickoff (₹6 lakh); 40% at user acceptance (₹8 lakh); 30% at go-live (₹6 lakh). Separately approved after the pilot gate.
Total through first production release	**₹35 lakh**	**Indicative fixed fees; not a commitment to proceed beyond each approval gate.**

Appendix A3 — Operating cost and value calculation

This appendix makes the recurring-cost and value assumptions explicit. All figures below are preliminary estimates, not vendor quotations or measured client outcomes.

3. Estimated recurring cost

Cost item	Indicative estimate	Included assumptions / exclusions
Application hosting, database/storage, backups and monitoring	**₹25,000–₹65,000 per month**	Modest CPU-based deployment; no GPU or paid LLM service. Depends on availability, retention and traffic requirements.
Ongoing engineering support	**₹1.0–₹1.5 lakh per month, separately priced**	Depends on SLA, response times, monitoring duties and included engineering hours. Not included in the ₹35 lakh implementation fee.

Excluded unless agreed: GST, StruMIS/PLM licence upgrades, new hardware or client-dedicated infrastructure, vendor charges, major scope changes and the separate support retainer. If discovery exposes materially incomplete data or proprietary integration work, re-scope before proceeding.

4. Value calculation — sensitivity, not forecast

The client described capacity of approximately 80,000 tonnes per year; actual throughput and the precise waste denominator are not yet verified. The following is only a simplified sensitivity example.

80,000 tonnes × 1 percentage point = 800 tonnes of material-equivalent change per year

If, purely for illustration, the net avoidable cost after scrap credits and incremental handling were ₹10,000 per tonne, the indicative gross material value would be 800 × ₹10,000 = ₹80 lakh per year, before hosting, support and implementation costs. This is not a savings forecast or a promise of payback. Finance must first approve the definition of material utilisation and the conversion from yield change to tonnes avoided for a comparable production output.

Use this ROI method after validation:

Step	Calculation	Control
1	Measure material input, required part output, retained offcuts and scrap for matched jobs.	Use the same denominator and classify material by family/job mix.
2	Estimate avoided input / recovered material only where the baseline and pilot are comparable.	Do not count an offcut once as saved material and again as scrap avoided.
3	Net benefit = verified purchase cost avoided + other verified savings − extra handling − scrap-credit changes − service/hosting/support cost.	Cost Control approves inputs and accounting treatment.
4	Payback = implementation cost ÷ annualised net benefit, only where net benefit is positive and repeatable.	Do not quote payback before the pilot has measured real outcomes.

Commercial decision gate: approve discovery first. Authorise the pilot only after data completeness, baseline measurement, manufacturing constraints and finance-approved value logic are reviewed. Approve production integration only if the pilot demonstrates safe and repeatable improvement with a positive business case.